

Problem

Find the time functions that have the following Laplace transforms:

(a) $F(s) = 10 + \frac{s^2 + 1}{s^2 + 4}$

(b) $G(s) = \frac{e^{-s} + 4e^{-2s}}{s^2 + 6s + 8}$

(c) $H(s) = \frac{(s + 1)e^{-2s}}{s(s + 3)(s + 4)}$

Step-by-step solution

Step 1 of 5

(a)

The function $F(s)$ is,

$$\begin{aligned} F(s) &= 10 + \frac{s^2 + 1}{s^2 + 4} \\ &= 10 + \frac{s^2 + 1 + 3 - 3}{s^2 + 4} \\ &= 10 + \frac{s^2 + 4 - 3}{s^2 + 4} \\ &= 10 + \frac{s^2 + 4}{s^2 + 4} - \frac{3}{s^2 + 4} \\ &= 10 + 1 - \frac{3}{s^2 + 4} \\ &= 11 - \frac{3}{s^2 + 2^2} \\ &= 11 - \frac{3}{2} \left(\frac{2}{s^2 + 2^2} \right) \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} L^{-1}[F(s)] &= L^{-1} \left[11 - \frac{3}{2} \left(\frac{2}{s^2 + 2^2} \right) \right] \\ f(t) &= 11\delta(t) - \frac{3}{2} \sin 2t \end{aligned}$$

Therefore, the time function of $F(s)$ is $\boxed{11\delta(t) - \frac{3}{2} \sin 2t}$.

Step 2 of 5

(b)

The function $G(s)$ is,

$$\begin{aligned} G(s) &= \frac{e^{-s} + 4e^{-2s}}{s^2 + 6s + 8} \\ &= \frac{e^{-s} + 4e^{-2s}}{s^2 + 6s + 8 + 1 - 1} \\ &= \frac{e^{-s} + 4e^{-2s}}{s^2 + 6s + 9 - 1} \\ &= \frac{e^{-s} + 4e^{-2s}}{(s+3)^2 - 1^2} \\ &= \frac{e^{-s}}{(s+3)^2 - 1^2} + 4 \frac{e^{-2s}}{(s+3)^2 - 1^2} \end{aligned}$$

Apply inverse Laplace transform.

$$L^{-1}[G(s)] = L^{-1}\left[\frac{e^{-s}}{(s+3)^2 - 1^2} + 4 \frac{e^{-2s}}{(s+3)^2 - 1^2}\right]$$

$$g(t) = e^{-3(t-1)} \sinh(t-1)u(t-1) + 4e^{-3(t-2)} \sinh(t-2)u(t-2)$$

Therefore, the time function of $G(s)$ is

$$\boxed{e^{-3(t-1)} \sinh(t-1)u(t-1) + 4e^{-3(t-2)} \sinh(t-2)u(t-2)}.$$

Step 3 of 5

(c)

The function $H(s)$ is,

$$\begin{aligned} H(s) &= \frac{(s+1)e^{-2s}}{s(s+3)(s+4)} \\ \frac{H(s)}{e^{-2s}} &= \frac{(s+1)}{s(s+3)(s+4)} \end{aligned}$$

Apply partial fractions.

$$\begin{aligned} \frac{(s+1)}{s(s+3)(s+4)} &= \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s+4} \\ \frac{(s+1)}{s(s+3)(s+4)} &= \frac{A(s+3)(s+4) + B(s)(s+4) + Cs(s+3)}{s(s+3)(s+4)} \\ (s+1) &= A(s+3)(s+4) + B(s)(s+4) + Cs(s+3) \dots\dots (1) \end{aligned}$$

Step 4 of 5

Substitute 0 for s in equation (1).

$$(0+1) = A(0+3)(0+4) + B(0)(0+4) + C(0)(0+3)$$

$$1 = A(3)(4)$$

$$A = \frac{1}{12}$$

Substitute -3 for s in equation (1).

$$(-3+1) = A(-3+3)(-3+4) + B(-3)(-3+4) + C(-3)(-3+3)$$

$$-2 = B(-3)(1)$$

$$B = \frac{2}{3}$$

Step 5 of 5

Substitute -4 for s in equation (1).

$$(-4+1) = A(-4+3)(-4+4) + B(-4)(-4+4) + C(-4)(-4+3)$$

$$-3 = C(-4)(-1)$$

$$C = -\frac{3}{4}$$

Therefore,

$$\frac{H(s)}{e^{-2s}} = \frac{1}{12} \frac{1}{s} + \frac{2}{3} \frac{1}{s+3} + \left(-\frac{3}{4}\right) \frac{1}{s+4}$$

$$H(s) = \frac{1}{12} \left[\frac{e^{-2s}}{s} \right] + \frac{2}{3} \left[\frac{e^{-2s}}{s+3} \right] - \frac{3}{4} \left[\frac{e^{-2s}}{s+4} \right]$$

Apply inverse Laplace transform.

$$L^{-1}[H(s)] = L^{-1} \left[\frac{1}{12} \left[\frac{e^{-2s}}{s} \right] + \frac{2}{3} \left[\frac{e^{-2s}}{s+3} \right] - \frac{3}{4} \left[\frac{e^{-2s}}{s+4} \right] \right]$$

$$h(t) = \frac{1}{12} u(t-2) + \frac{2}{3} e^{-3(t-2)} u(t-2) - \frac{3}{4} e^{-4(t-2)} u(t-2)$$

Therefore, the time function of $H(s)$ is

$$\boxed{\frac{1}{12} u(t-2) + \frac{2}{3} e^{-3(t-2)} u(t-2) - \frac{3}{4} e^{-4(t-2)} u(t-2)}.$$